



The Midnight Mango Thief Adventure!

Grade 1 STEM Lesson Guide: Make a Buzzer Alarm

This revised primary STEM instruction manual simplifies hardware loop logic, switching behaviors, and electrical safety boundaries into child-friendly, high-energy language suitable for Grade 1 students.

SLIDE 1: WELCOME, SUPER ENGINEERS!

What's on the screen: A bright, colorful title page with big letters: "Make a Buzzer Alarm! Catch the Mango Thief!"

WHAT TO SAY:

"Hello, super inventors! Welcome to our science lab! Today, we are going to get a special superpower. We are going to learn how to use electricity to make things work and solve a real-life mystery! Are you ready? Let's go!"

SLIDE 2: THE STORY MISSION

What's on the screen: A cartoon of our friend Ayaan standing next to a huge, beautiful tree full of shiny yellow Alphonso mangoes, with a cheeky little monkey hiding behind the leaves.

WHAT TO SAY:

"Look at the screen! This is our friend Ayaan. His grandfather has a gorgeous backyard tree full of yummy, sweet yellow mangoes in Ratnagiri. But oh no! Every single night at midnight, a sneaky, mischievous little monkey climbs down the branches and takes a giant bite out of the very best fruits! Ayaan cannot stay awake all night long to guard them. Can we use our technology skills to build a secret electronic sound alarm trap to catch the thief? Let's help Ayaan protect his mangoes!"

SLIDE 3: WHAT IS A CIRCUIT?

What's on the screen: Drawings of tiny, energetic "Electricity Ants" wearing capes.

- **Open Circuit:** A broken path where the ants hit a gap and freeze. No sound!
- **Closed Circuit:** A perfect loop where the ants march smoothly to the buzzer and back.

WHAT TO SAY:

"To make our alarm system work, we need to understand a secret path called a **CIRCUIT**

. Let's imagine that inside every wire, there are billions of tiny, invisible, energetic 'Electricity Ants'! These ants love to run and carry power. But they have one major rule: they will only start marching if they have a perfect, unbroken circular track to walk on! If there's a gap or a broken line, it's an

OPEN CIRCUIT

—the ants freeze because they can't jump! But if we connect the line perfectly, it becomes a

CLOSED CIRCUIT

. The ants march all the way through the sound maker and back home!"

SLIDE 4: EVERYDAY CIRCLE TRACKS

What's on the screen: Real-world examples of loops from daily life in India.

- **The Metro Train:** A full circle track keeps the train moving smoothly through the city.
- **Festival Rice Lights:** Beautiful twinkling lights on the balcony. If a window pinches and snaps one tiny wire, the whole strand goes dark!

WHAT TO SAY:

"We see these ant tracks everywhere around us! Think of the busy Metro trains in Delhi, Mumbai, or Bangalore. If the circular track is entirely built and continuous, the train runs smoothly from station to station. If a section is removed, the track breaks and the train stops instantly! It's just like the beautiful festive rice lights we hang up for Diwali or Eid. If just one tiny wire gets clipped by a window edge, the whole sequence goes dark because the ant pathway is open!"

SLIDE 5: MEET THE BUZZER!

What's on the screen: A happy little round piezo buzzer component making sound waves.

- **Doorbell:** Sings 'Ding-Dong' when a guest arrives!
- **School Bell:** Rings loudly when it's recess time!
- **Microwave:** Beeps to tell you your food is ready!

WHAT TO SAY:

"Let's discover our alarm's mouth—the piezo buzzer! This little component takes the fast running speed of our electricity ants and turns it instantly into clear sound waves. Where have you heard this sound today? Yes! When a delivery person pushes your apartment doorbell button and it sings 'Ding-Dong'! Or when the school hallway bell loudly echoes across classrooms to tell you it's time to play!"

SLIDE 6: THE SECRET TRIGGER

What's on the screen: A pressure doormat switch. It shows two hidden metal pads inside that are kept apart until someone steps on them.

WHAT TO SAY:

"How will our alarm know the monkey is there? We will use a secret

PRESSURE SWITCH

trigger! Normally, the switch hangs out in an open state—holding the wires apart so the circuit stays completely silent. The ants are waiting. But the moment the mango thief steps his foot down on our garden doormat plate, his mechanical weight forces the hidden metal pads inside to touch, closing the gap instantly. The pathway is complete!"

SLIDE 7: STEP 1 - CONNECT THE POWER

What's on the screen: A heavy-duty yellow 9V battery and a buzzer on a digital workbench.

Action: Drag the **red wire** from the battery to the buzzer's **red terminal**.

WHAT TO SAY:

"Alright, mini engineers, stretch your magic fingers out! Look at our digital workbench screen. We have a yellow 9V battery pack. Touch the positive red wire line, pull it gently across the screen, and snap it directly onto the red leg of our buzzer. Outstanding! The power supply is locked into position and the electricity ants are lining up to march!"

SLIDE 8: STEP 2 - ADD THE MAT SWITCH

What's on the screen: Routing the ground line.

Path: [Buzzer Black Leg] → [Switch Inlet] → [Switch Outlet] → [Battery Black Side]

WHAT TO SAY:

"Now let's complete the whole circle track! Take the black wire from the buzzer, connect it to our secret mat switch, and run the other side back to the black side of the battery. Super Engineer, our loop is almost complete!"

SLIDE 9: STEP 3 - TEST THE ALARM!

What's on the screen: A virtual stone dropping right onto the garden doormat switch, making a giant sound wave graphic: **BZZZZZZ!**

WHAT TO SAY:

"It's midnight in the orchard! Let's test our trap before the monkey arrives. Pick up that large virtual testing stone from your screen panel and drop it right on top of our garden doormat switch. Look! The mat squishes down, the gap closes, the circuit completes, and listen—

BZZZZZZ!

It works beautifully! Shabash! Good job!"

SLIDE 10: FIX-IT PUZZLE 1 (WRONG WIRES!)

What's on the screen: The wires are connected upside down, and the buzzer is totally quiet.

WHAT TO SAY:

"Oops! The monkey is approaching, but our buzzer isn't ringing! Let's check our work. Oh look—the wires are connected backwards! Electricity ants from our toy batteries are like cars on a one-way street; they can only march in one specific direction. If the red and black wires are swapped, the ants get confused and freeze. Let's flip the wires around to help them march the right way!"

SLIDE 11: FIX-IT PUZZLE 2 (THE BLOCKING BLOCK)

What's on the screen: A plastic toy building block blocking the wire vs. a shiny metal safety pin.

- **Insulators (Plastic):** Act like giant walls that freeze the electricity ants.
- **Conductors (Metal):** Act like smooth, paved open highways where ants love to sprint!

WHAT TO SAY:

"Oh no, another puzzle! There is a plastic toy block stuck in our path, and the buzzer won't ring! Plastic is an insulator—it acts like a giant brick wall that completely blocks our path. The ants hit it and can't move forward! Can you spot the metal safety pin on your screen? Metals are conductors, which means they are like smooth, paved highways for electricity. Let's swap out that plastic chunk for the metal pin to open up the highway!"

SLIDE 12: THE ELECTRICAL SAFETY RULES

What's on the screen: Three pictures side-by-side:

- A small toy AA battery pack **SAFE**
- A household wall plug socket **NEVER TOUCH!**
- A battery with a warning sign **ASK AN ADULT**

WHAT TO SAY:

"Before we celebrate, let's learn how to be safe superstar scientists. Look closely at these pictures. Small toy batteries (like AA or 9V batteries) have gentle power that is perfectly safe for young inventors like us. But look at the big wall sockets in your house. Those carry giant, dangerous high-voltage power! You must

NEVER EVER

touch wall outlets or stick anything into them. Always ask a grown-up before doing any science experiment!"

SLIDE 13: THE STEM SAFETY SCIENTIST OATH

What's on the screen: A bright golden badge that reads: *Official Junior Lab Assistant.*

WHAT TO SAY:

"Raise your right hand along with me, superstar inventors! Let's say our sacred safety pledge together:

'I PROMISE TO ALWAYS USE SAFE TOY BATTERIES FOR MY INVENTIONS. I WILL NEVER, EVER TOUCH BIG HOUSEHOLD WALL PLUGS, AND I WILL ALWAYS ASK MY PARENTS OR TEACHER FIRST!'

Outstanding! You are now an officially certified Planrs Academy Junior Lab Assistant!"

SLIDES 14 & 15: HOMEWORK & WRAP-UP

What's on the screen: Ayaan cheering with a huge basket of perfectly safe mangoes, and the cheeky monkey waving goodbye as it runs back into the jungle.

WHAT TO SAY:

"Hooray! Ayaan's orchard is perfectly safe now, thanks to your incredible wiring skills! The sneaky monkey heard our loud alarm and safely ran away. Your fun homework today is to look around your kitchen and living room and spot just one device that uses a buzzer or makes a beep. Draw it or write it down in your notebook to share next time. Thank you, super engineers, and keep inventing!"